COM-480 Milestone 2 - DataRizz

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1 Project Goal

The goal of our project is to provide insights into how inequalities have been reflected in the Olympic Games over time. We explore aspects such as gender, GDP, disciplines, geopolitics, and sociocultural contexts through our visualisations. We aim to present these visualisations through an interactive and user-friendly website.

2 Outline

Below, we present the proposed structure of visualisations for our website.

Where Do the Medal Winners Come From?

Understanding how Olympic medals are distributed across disciplines and regions can provide insight into global performance patterns and how they have evolved over time. The first section shows the distribution of gold, silver and bronze medals across the top 10 disciplines. This is displayed year by year, and broken down by winter/summer and continent. A time slider below the figure allows users to explore how this distribution has changed over time. The plot, implemented as a Sankey diagram, also shows the relative contribution of each continent. Users can select or deselect continents to display the corresponding delegation countries for a given year.

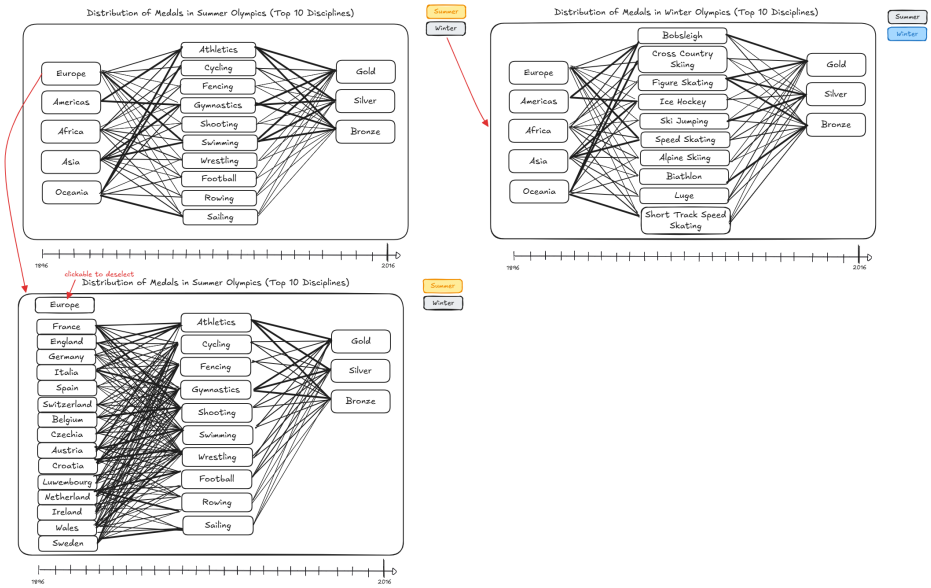

Figure 1: Medals around the world

How has gender representation evolved across Olympic sports and disciplines over time?

Women joined the Olympic Games in their second edition. However, gender parity has taken more than a century to approach. To address this research question, we represent the various sports and their disciplines using a sunburst plot for both men and women. The color scale is lighter when the proportion of athletes of a given gender is lower and darker when it is higher. This visualization highlights the evolution of the proportion of men and women in each discipline over the years. A time slider below the visualization allows users to explore different years and can be paused at any moment. Users can also choose between Summer and Winter Olympic Games. When hovering over a sport or discipline, the exact percentage is displayed. In addition, we include a pie chart that illustrates the overall proportion of men and women across all disciplines.

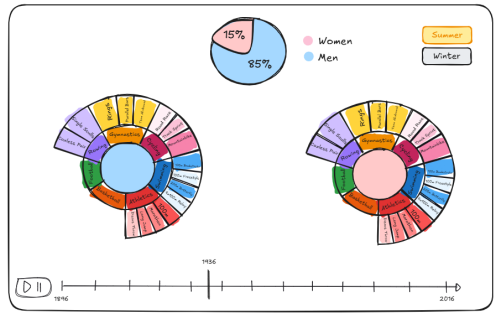

Figure 2: Gender evolution with pie chart and sunbursts

How do geopolitical conflicts affect the Olympic Games?

Major geopolitical conflicts, such as the World Wars, the Cold War, and Olympic boycotts, have shaped participation and performance in the Olympic Games. To address how geopolitical conflicts affect the Olympic Games, we first present a map highlighting the countries involved, along with key geopolitical events that have influenced the evolution of the Games. This provides contextual insights for the user. For example, the 1976 Montreal Olympic boycott saw 28 countries, mostly African, withdraw from the Games in protest against the participation of New Zealand, which was perceived as supporting apartheid. We also present two additional visualizations: when the selected year corresponds to a conflict, one shows the number of medals won by each country involved in the conflict, while the other illustrates the number of countries participating in the Olympic Games around that year.

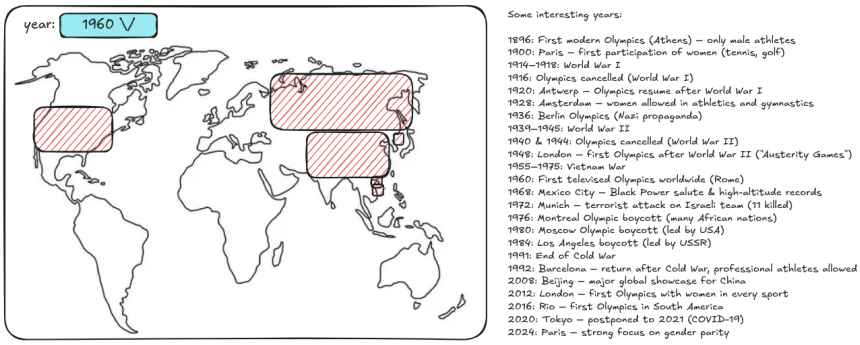

Figure 3: Geopolitical conflicts map

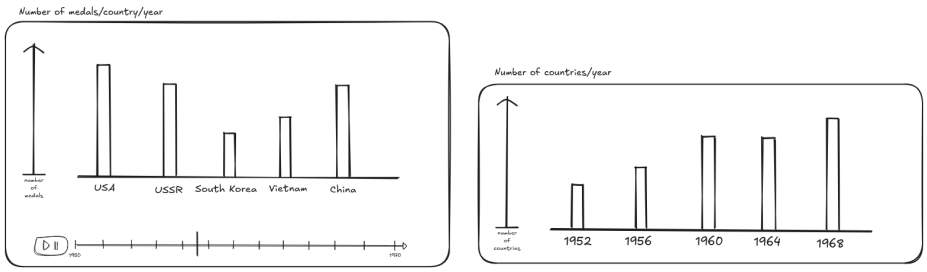

Figure 4: Geopolitical conflicts graphs

What Would a Fair Distribution of Olympic Medals Look Like?

The geopolitical and sociocultural contexts affect the resources available to delegations at each Olympic Games, in terms of population size and GDP. These inequalities influence the outcome of the competitions. Out of curiosity, what would a fair distribution of Olympic medals look like? Which countries would have a better or worse success rate? The Figure 5 is an interactive visualisation applying course concepts such as filtering by season, year and country, aggregation using the fair medal plot and linked views for comparison. The user can pick the season (summer or winter), the year and two countries to be compared. We calculate the fair number of gold, silver and bronze medals as follows:

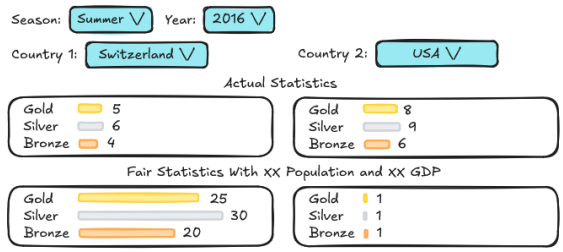

Figure 5: Fair medal distribution comparison

$$\# \text{ Fair medals} = \# \text{ Actual medals} \times \frac{\text{Mean population}}{\text{Country's population}} \times \frac{\text{Mean GDP}}{\text{Country's GDP}}$$

3 Tools

Our interactive website will be built using standard web technologies introduced throughout the course. HTML and CSS will be used to structure and style the website. We were introduced to these tools in Week 1, and they will enable us to create a clear, user-friendly layout for our visualisations. JavaScript will be used to implement interactivity and dynamic content (Weeks 2 and 3). It will handle user inputs, updates to the visualisations and coordination between the interface's different components. `D3.js` will be our main visualisation library (Weeks 4 and 5). It will be used to create interactive, customised visualisations, such as bar charts that evolve over time and zoomable sunburst diagrams. The concepts from Week 5 on interactive views will inform the design of user interactions (e.g. filtering, zooming and comparisons), while the concepts from Week 6 on colour perception will influence our visual encoding choices to ensure readability and effective communication. In addition, we will rely on the material from Week 8 on map design to implement the geographic visualisations. Finally, GitHub Pages will be used to host the project, allowing us to deploy and share our interactive website. Furthermore, we used Excalidraw to create the sketches for this report.

4 Additional Ideas

If time permits and the ideas are feasible to implement, we would like to include the following features in our final website:

1. A first idea would be to create an interactive visualisation showing the body types of medallists. Users would be able to filter by season, year, gender, sport and discipline. The visualisation in Figure 6 would display a podium with the body types of the winners, showing their height, weight, name and delegation. To implement this, we could render the scene in 3D using `Three.js`, load a human model (MakeHuman), apply morph targets to adjust weight and supposed BMI (Blender), and scale the model to the correct height.

2. Another idea would be to add interesting facts to the "Where Do the Medal Winners Come From?" visualisation to help explain some of the trends observed in the data. These facts would provide additional context and make the visualisation easier to understand. For each year, one or two key insights explaining the width of certain flows in the Sankey diagram would be highlighted and presented below the plot, allowing users to better interpret the data. The visualisation in Figure 7 represents this idea.

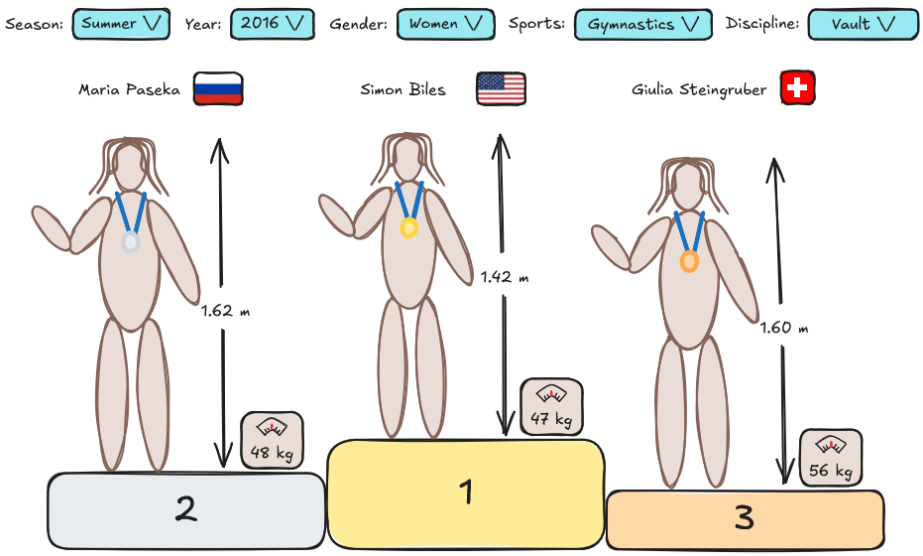

Figure 6: Podium of body types

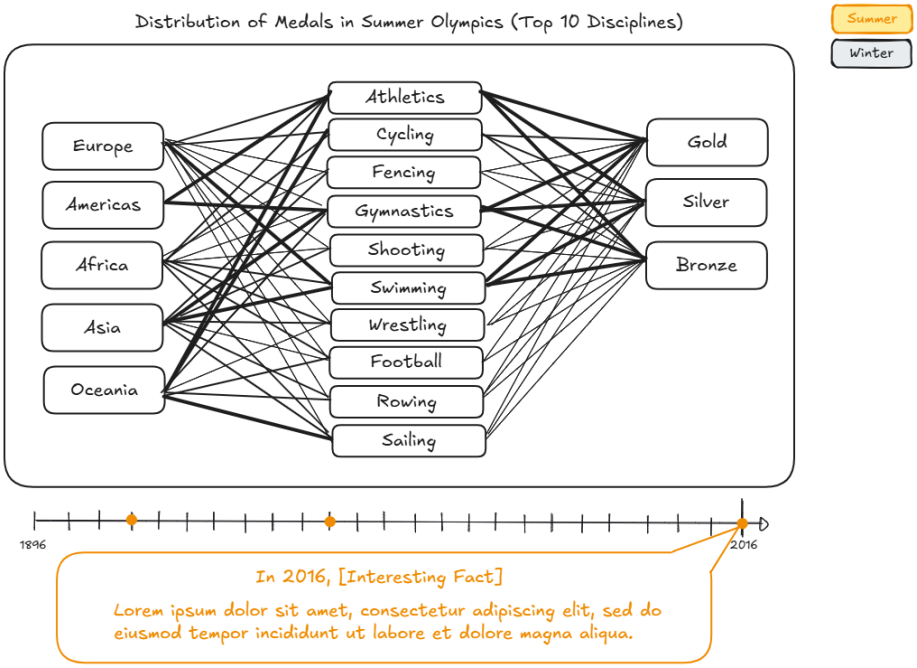

Figure 7: First visualisation with interesting facts